When Disagreement Labels Do Not Identify F1 Rankings

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Abstract. Agreement labels cancel from an accuracy comparison but can change an F1 ranking. We specialize partial identification to an explicit audit after labeling disagreements: two agreement counts give exact bounds, certifying a ranking or a choice within a specified F1 tolerance under every completion. Across 3,200 frozen model pairs, shared completions tighten bounds and a 0.01 tolerance reduces labeling costs. A zero-loss (weak-winner) control attributes most savings to positive tolerance, not ties. Guarantees concern the evaluated batch, not future data.

1 What can disagreement labels decide?

Suppose two fixed classifiers disagree on three of ten items. One predicts positive on all three, the other negative; only one of these labels is positive. They agree positive on three other items and negative on four. Even knowing that exactly four of the ten labels are positive does not settle their F1 ranking. Allocating the three remaining positives to negative agreements gives F1 scores $(1/5, 0)$; allocating them to positive agreements gives $(4/5, 6/7)$. The same disagreement labels and prevalence support opposite winners.

Given predictions, disagreement labels and optional prevalence, we reduce unknown agreement labels to two counts. Their exact F1 bounds certify a choice at the requested tolerance or require agreement queries, preserving a shared label completion.

Discordant-pair evaluation by Musgrove et al. [1] uses prevalence and baseline sensitivity/specificity assumed to transfer to the validation data. We have no such baseline-performance information; the example above therefore does not refute their method. Active F-measure estimation [2] and Okanovic et al.'s Bayesian, accuracy-based best-or-near-best selection [3] address label efficiency without our deterministic all-completion F1 guarantee. F1 batch dependence [4] and single-model missing-label bounds [5] are established. We give an explicit specialization of joint identification frameworks [6, 7, 8].

2 An explicit audit criterion

Fix binary predictions $q, c \in \{0,1\}^n$ and correct unknown labels. Let a, b count disagreements $(q, c) = (1, 0), (0, 1)$; n_{00}, n_{11} count agreements. After all $D = a+b$ disagreement labels are revealed, let u, v be their respective positive counts.

Unknown positive counts on $(0,0)$ and $(1,1)$ agreements are s, t, respectively. Write

$$P = u + v + s + t, \quad Q = a + n_{11}, \quad C = b + n_{11}, \quad d = u - v.$$

P is the true positive count, while Q, C are observable predicted counts. Positive-class F1 is $2\mathrm{TP}/(P + \text{predicted positives})$, with $0/0 = 0$. The paired difference is

$$\Delta(s,t) = F_1(q) - F_1(c) = \frac{2(t+u)}{P+Q} - \frac{2(t+v)}{P+C}. \tag{1}$$

By contrast, accuracy difference $(2d + b - a)/n$ is already determined; balanced accuracy difference $[nd + (b-a)P]/[2P(n-P)]$ is determined if $0 < P < n$ is known, and undefined at $P = 0, n$.

Sharp sign criterion (specialization). Set $e = d + b - a$, $K = d(u + v + n_{11}) + ub - va$. Then Δ has the sign of

$$S(s,t) = K + ds + et. \tag{2}$$

Without prevalence information, agreement queries leave an integer rectangle $s \in [s_-, s_+]$, $t \in [t_-, t_+]$, initially $[0, n_{00}] \times [0, n_{11}]$. Its exact numerator bounds are

$$\begin{aligned} S_- &= K + \min(ds_-, ds_+) + \min(et_-, et_+), \\ S_+ &= K + \max(ds_-, ds_+) + \max(et_-, et_+). \end{aligned} \tag{3}$$

Both strict rankings are possible iff $S_- < 0 < S_+$; $S_- > 0$ or $S_+ < 0$ identifies a strict winner, and $S_- = S_+ = 0$ an exact tie. Other cases allow a tie and one winner.

Proof. Cross multiplication gives $\Delta(P+Q)(P+C)/2 = dP + (C-Q)t + uC - vQ = K + ds + et$. The denominator is positive unless $P = 0$ and a predictor has no positives; then both F1 scores and S are zero. Every integer count in the rectangle is attainable. An affine function reaches its extrema at its corners, proving the sign claims. □

Equal-count corollary. If $Q = C = R > 0$, then $\Delta = 2d/(P + R)$: disagreement labels identify the sign without knowing P. If $R = 0$, both scores vanish. Magnitude can remain unknown.

Exact magnitude bounds. For fixed $P > 0$, the difference is affine in t:

$$f_P(t) = \frac{2u}{P+Q} - \frac{2v}{P+C} + \frac{2(C-Q)t}{(P+Q)(P+C)}. \tag{4}$$

The feasible endpoints are $\ell_P = \max(t_-, P - u - v - s_+)$ and $h_P = \min(t_+, P - u - v - s_-)$. If $\ell_P \le h_P$, evaluating f_P at both endpoints gives its exact hull; otherwise that P is infeasible. At $P = 0$ both scores are zero. With unknown P, enumerate feasible integer counts P and take the smallest and largest of these endpoints, yielding the exact interval hull $[L, U]$ in $O(n)$ time. This includes $P =$

n. The sign proof does not place unknown-P magnitude extrema at rectangle corners.

A choice with bounded regret. For absolute F1 tolerance $\varepsilon \geq 0$, the worst losses from selecting q and c are $R_q = \max(0, -L)$ and $R_c = \max(0, U)$. Choose the smaller, breaking ties in favor of q, and stop iff

$$\min(R_q, R_c) \leq \varepsilon \quad \Longleftrightarrow \quad L \geq -\varepsilon \text{ or } U \leq \varepsilon. \tag{5}$$

This is Kuboyama's tolerance relaxation and minimum worst-case excess choice [6, §3, §5] on the F1 hull. These losses prove the batch guarantee. It does not require $[L, U] \subseteq [-\varepsilon, \varepsilon]$. At $\varepsilon = 0$ it allows a weak winner before distinguishing a strict win from an exact tie. Failure means $L < -\varepsilon$ and $U > \varepsilon$: opposite compatible advantages exceeding the tolerance. The inclusive material-reversal condition $L \leq -\varepsilon$, $U \geq \varepsilon$ differs only at the boundaries.

Relation to general identification frameworks. Kuboyama [6], Theorem 1, Proposition 1, §4.5 and §6.2, gives joint all-completion certificates, agreement relevance and tolerance for selective prediction. For $H \in \{Q, C\}$ and known P with $P + H > 0$, $1 - F_1 = (\mathrm{FP} + \mathrm{FN})/(P + H)$ is an additive weighted error. Sharp bounds must nevertheless retain $\sum_i y_i = P$; his unrestricted-label extrema cannot simply be reused under this constraint. With unknown P, denominators vary across completions. Our contribution is the explicit F1 audit, including degeneracies, not general certificate, joint-comparison or tolerance principles.

TDC [7], Theorem 1, gives sharp certificates for fixed additive-loss contrasts over unrestricted completions. With $P > 0$ fixed,

$$\Delta = \sum_i y_i \left[\frac{2q_i}{P + Q} - \frac{2c_i}{P + C} \right]. \tag{6}$$

Imposing $\sum_i y_i = P$ is essential for sharpness; TDC's §4.5 already groups items by prediction signature, constrains their positive counts and enumerates prevalence. Equations (2)–(4) carry this reduction to F1.

Polo et al. [8], Eq. (1.1) and §3.2, also cover fixed-P F1: use uniform batch indices, distinguish audited items and fix the label marginal. For (6), the remaining fractional labels have an integer sum. Every vertex is integral: two fractional entries permit a sum-preserving perturbation; one cannot sum to an integer. Hence linear extrema equal our completion extrema. Our contribution is the explicit audit and its consequences.

Separate bounds form $[\min F_1(q) - \max F_1(c), \max F_1(q) - \min F_1(c)]$. The feasible set is shared, but the optimizers need not agree. Marginals at fixed P are affine in t, so enumeration is exact.

3 Retrospective evaluation

Frozen cases and units. We reuse TDC's prediction artifact, release 1.1.6 [9], without refitting models. Each of 36 cells contains 500 targets, a validation-selected quantum-kernel predictor, and 115 classical configurations. Twenty cells

cover TableShift's [10] BRFSS diabetes and ACS FoodStamps (five seeds, SVC and Gaussian-process classification); 16 cover eight EMBER [11], network ToN-IoT [12] and UNSW-NB15 [13] scenarios with both classifiers. ToN-IoT uses the original `train_test_network.csv` scanning-task variant, not sensor telemetry or a NetFlow re-export. We collect no new data. Deduplicating identical classical predictions within cells yields 3,200 pairs. The criterion applies to arbitrary fixed binary predictions.

The pilot (eight cells, 801 pairs) preceded label inspection in this study; the reserved 28 cells (2,399 pairs) are an analytical holdout of existing evidence. Tolerance, separate-bound and validation analyses are retrospective extensions.

Fractions average equally across cells; costs average orders, pairs within cells, then eligible cells. Shared targets and overlapping subsamples preclude independent-observation inference; summaries are descriptive. Known P is an exact information oracle. Initial intervals retain $0 < P < n$, the original balanced-accuracy domain; query cohorts and stops allow $P = 0, n$.

Acquisition and comparators. We retain three agreement orders: uniform; alternating shuffled agreement strata; and decreasing $|d|$ or $|e|$ (magnitude, shuffled ties). Seeds 0–31 share permutations across policies and objectives. Orders use predictions and disagreement labels only; queries shrink the count intervals. Magnitude applies Kuboyama's static range order [6, Eq. (10)] to S: it minimizes numerator width, not necessarily F1 query cost. We compare strict-sign/exact-tie identification, a weak winner ($\varepsilon = 0$), and $\varepsilon = 0.01$ on the same trajectories. Full labels verify regret; they never select pairs, queries or stopping decisions.

We select single classical comparators by minimum Hamming distance, with original-order ties, or maximum validation balanced accuracy, with ties broken lexicographically by configuration. All 115 validation scores were available; selection preceded the new target analysis and follows TDC's objective, not F1. Hamming proximity is a sensitivity analysis, not a deployment rule.

Mechanism. At known $P > 0$, interval width is exactly $2|C-Q|(h_P-\ell_P)/[(P+Q)(P+C)]$: predicted-count imbalance and freedom to allocate positives between agreement groups jointly drive ambiguity. Without P, initial numerator width is $|d|n_{00}+|e|n_{11}$. All 58 equal-count pairs identify the sign but retain nonzero mean magnitude uncertainty; disagreement count alone does not explain ambiguity.

Initial decisions. In the 28 reserved cells, the mean material-reversal fraction is 20.54% without P, and 10.52% with exact P. These are compatible opposite advantages of at least 0.01, not observed errors or universal frequencies. No pair lies on the 0.01 boundaries, so complements are certifiable at that tolerance.

Keeping dependence matters: in the reserved cells, paired bounds certify 79.46% of choices at $\varepsilon = 0.01$, versus 6.57% for separate bounds; with exact P, 89.48% versus 18.80%. Mean widths are respectively 0.173 versus 1.585, and 0.034 versus 1.005. This is an information ablation on identical feasible sets, not superiority over joint Kuboyama, TDC or Polo certificates.

Figure 1 shows the tolerance dependence: at 0.01, certification is 79.15% over all 36 cells, versus 79.46% in the 28 reserved cells. Hamming selection

leaves material ambiguity in 3/36 cells; validation selection leaves 6/36 (both have 11/36 strict reversals). These one-pair-per-cell designs answer a different question from the all-configuration pool.

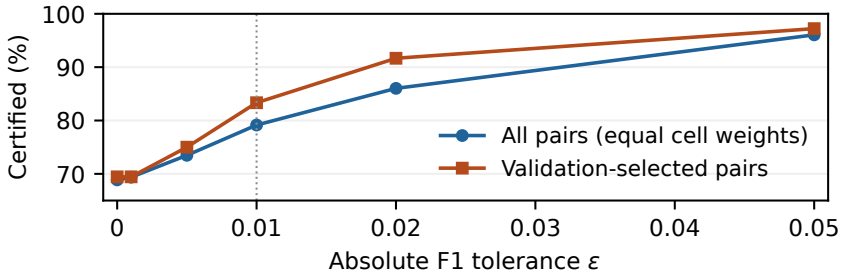


Fig. 1: Certification after disagreement labeling, unknown prevalence ($0 < P < n$). Equal-cell means over all 36 cells: 3,200 pairs, or one validation-selected pair per cell. Points are evaluated tolerances; lines guide the eye. Descriptive fractions, not confidence estimates.

Label costs. Table 1 includes already-decided pairs. Under magnitude ordering, a 0.01 tolerance reduces total labels from 121.16 to 74.59 (all pairs), and 92.57 to 40.29 (validation), versus 500 for full labeling. Zero-loss totals, 120.92 and 92.25, remain close to strict/exact costs: positive tolerance accounts for most savings.

		Initial	Additional		Total	
Cohort	Policy	D	Exact	ε	Exact	ε
All pairs	Uniform	54.5	88.21	35.58	142.70	90.07
All pairs	Alternating	54.5	80.64	31.29	135.13	85.78
All pairs	Magnitude	54.5	66.67	20.10	121.16	74.59
Validation	Uniform	30.7	89.07	25.56	119.79	56.28
Validation	Alternating	30.7	77.58	16.34	108.30	47.07
Validation	Magnitude	30.7	61.84	9.57	92.57	40.29

Table 1: Mean labels on common cohorts, 36 cells, 32 orders per policy. All pairs: 3,200 comparisons; validation: one per cell. Additional labels query agreements; $\varepsilon = 0.01$. Each batch has 500 items. Totals are calculated before rounding.

Costs vary across cells: under magnitude ordering and $\varepsilon = 0.01$, the all-pair cell means have minimum/median/maximum 14.0/77.0/188.4 labels. This is descriptive dispersion, not an uncertainty interval. The overall mean is 74.59 with equal cell weights, versus 82.05 with equal pair weights. In 15,466 of 307,200 dependent replays, the certified choice is the realized-worse model; maximum regret is 0.00927, within tolerance. The guarantee is (5); these are retrospective checks.

4 Use and limitations

Given a tolerance, an auditor computes $[L, U]$ from the current count bounds, then chooses by (5) or queries agreements and updates those bounds.

This specialization assumes fixed predictions and correct labels; its guarantee does not extend to populations, noisy labels, macro-F1, multiclass or families. Compatible extremes need not be probable; benchmark reuse limits breadth. Disagreement-first labeling is imposed. Costs are neither optimal-acquisition lower bounds nor additive across pairs. Statistical alternatives can trade assumptions or error probability for fewer labels.

Availability. Code, frozen inputs and results (v1.4.0): `https://doi.org/10.5281/zenodo.23647323`.

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